

EMIOS – Enterprise Migration Intelligence Operating System

Vision

Instead of asking *How many hours will migration take?*, we're answering *How should this enterprise evolve safely and profitably?*

Problem Statement

Enterprise migration assessments are often inaccurate because teams do not fully understand the legacy ecosystem.

Migration failures are commonly caused by:

- Hidden dependencies
- Undocumented business rules
- Tribal knowledge
- Unknown downstream impacts
- Poor migration sequencing
- Incomplete risk visibility

Organizations need an intelligent platform that understands enterprise landscapes, simulates migration outcomes, predicts risks, and recommends modernization strategies before migration begins.

What Makes EMIOS Different

Existing Tools	EMIOS
Static dependency scanners	Living Digital Twin that evolves as the system changes
Siloed risk reports	Collaborative multi-agent system where agents negotiate trade-offs
Consultant-driven estimates	Simulation-first planning with Monte Carlo confidence intervals
Black-box recommendations	Explainable AI that shows <i>why</i> a path was chosen
One-time assessments	Self-learning knowledge base that preserves organizational memory

Proposed Solution

EMIOS creates a Digital Twin of enterprise systems and uses AI agents to autonomously discover, analyze, simulate, and plan migration initiatives.

Core Modules (Full Platform)

Module	Purpose
Enterprise Digital Twin	Real-time mirror of systems, data flows, and business logic
Migration Intelligence Graph	Knowledge graph connecting applications, dependencies, business rules, and risks
Autonomous Discovery Agent	Crawls codebases, configs, and documentation to map the landscape without manual input
Multi-Agent Planning Engine	Orchestrates specialized agents that negotiate migration sequencing and resource allocation
Simulation Engine	Runs “What-If” scenarios against the Digital Twin before any real changes occur
Monte Carlo Risk Prediction	Generates probabilistic risk distributions with confidence intervals for timeline and cost
Business Impact Engine	Quantifies revenue, compliance, and operational impact of each migration path
Explainable Recommendation Engine	Translates agent decisions into human-readable rationale with visual evidence
Self-Learning Knowledge Base	Captures tribal knowledge and improves recommendations with every migration
Executive Decision Copilot	Natural language interface for C-level stakeholders to explore scenarios

Recommended MVP

Pre-built before hackathon: Digital Twin visualization framework, Neo4j schema, and React dashboard shell.

Built During Hackathon

Component	Agentic Behavior
Migration Intelligence Graph	Auto-populated from uploaded metadata
Discovery Agent	Autonomously identifies undocumented services by analyzing connection patterns; flags “orphan” components for human review
Dependency Agent	Detects circular dependencies and proposes decoupling strategies; escalates critical paths to Planner Agent
Risk Agent	Independently calculates failure probability per component; negotiates timeline adjustments with Planner Agent when risk exceeds 15% threshold
Planner Agent	Generates migration waves by balancing risk, dependency order, and business priority; accepts or rejects Risk Agent proposals
What-If Simulation Engine	Executes migration scenarios against the Digital Twin; surfaces cascading failures
Explainable Recommendation Dashboard	Shows <i>why</i> each recommendation was made citing specific dependencies, risk scores, and agent negotiation logs

Demo Narrative

1. Upload migration metadata (CSV/API dump).
2. Generate Enterprise Digital Twin in real time.
3. Visualize Migration Intelligence Graph watch agents discover hidden dependencies.
4. Run “What-If” migration simulations see cascading risk propagation.
5. Display business impact and probabilistic risk ranges.
6. Generate migration roadmap with explainable agent rationale.

Business Value

Metric	Before EMIOS	With EMIOS
Discovery of hidden dependencies	Manual, incomplete	Automated, continuous
Risk visibility	Qualitative, static	Quantified, simulated
Knowledge preservation	Tribal, ephemeral	Structured, self-learning

Innovation Highlights

- Digital Twin for live enterprise migration modeling
- Knowledge Graph-driven planning with relationship-aware intelligence
- Agentic AI collaboration agents negotiate, escalate, and resolve conflicts autonomously
- Business-aware dependency analysis linking technical debt to revenue impact
- Explainable AI with full decision audit trails
- Self-learning migration intelligence that improves with every engagement

Tech Stack

Layer	Technology
Frontend	React + Material UI
Backend API	FastAPI
Agent Orchestration	LangGraph
LLM	Azure OpenAI / GPT-4o
Graph Database	Neo4j
Vector Store	Qdrant
Observability	Langfuse
Deployment	Docker + Azure

Story

EMIOS is not another migration estimation tool. It is a strategic modernization platform that enables enterprises to understand, simulate, and optimize migration decisions before execution begins.

We believe migration planning should be intelligent, transparent, and continuous not a one-time guess that expires the moment the first server is touched.